AQUAH: Automatic Quantification and Unified Agent in Hydrology

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AQUAH

Automatic Quantification and Unified Agent in Hydrology

HyDROS Lab

Start Flash Flood Analysis with AQUAH

AI-Powered
Advanced machine learning

Real-time Analysis
Live flood monitoring

Spatial Data
Interactive mapping

Hydrologic Modeling Faces Critical Accessibility Barriers



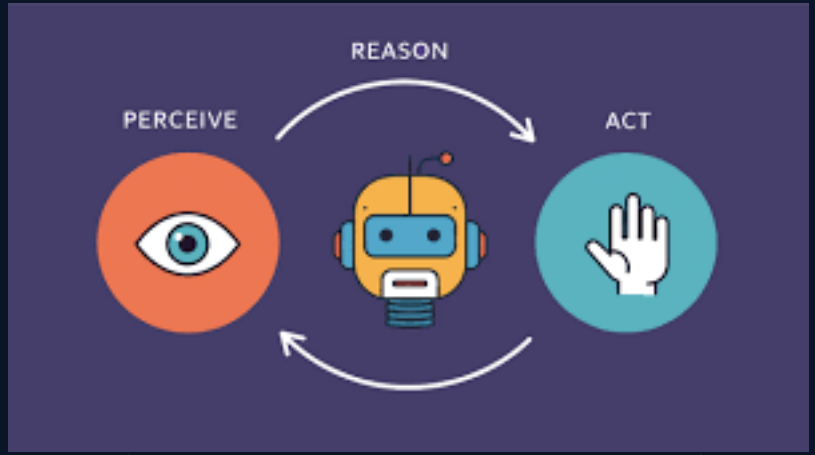
Current hydrologic simulation tools impose steep barriers that restrict their use to domain experts. Users must manually download terrain and forcing data, configure complex physics-based models, and interpret outputs—tasks requiring both specialized domain knowledge and programming expertise. Model setup alone can consume days or weeks, making rapid-response applications infeasible. This fragmentation prevents non-experts, emergency managers, and decision-makers from accessing advanced water resource modeling capabilities precisely when climate change intensifies flood risks and water scarcity challenges.

Key Challenges

- Manual data acquisition and preprocessing
- Steep learning curve for model configuration
- Time-consuming parameter calibration
- Results interpretation requires years of experience
- Limited accessibility for rapid-response scenarios

AQUAH Transforms Natural Language into Complete Hydrologic Simulations

AQUAH represents the first end-to-end language-based agent designed specifically for hydrologic modeling. Starting from a simple natural language prompt such as "simulate floods for the Little Bighorn basin from 2020 to 2022," AQUAH autonomously retrieves required terrain and forcing data, configures the hydrologic model, executes the simulation, and generates a self-contained PDF report. This breakthrough leverages vision-enabled large language models that interpret maps and rasters on the fly, replacing expert-driven decisions with reliable data-driven automation.



Core Capabilities

- Natural language interface for simulation requests
- Autonomous data retrieval and preprocessing
- Automated model configuration and execution
- Vision-based map and raster interpretation
- Self-contained report generation

System Architecture Integrates Six Modular Components

AQUAH's architecture bridges natural language interaction with Earth observation data, geospatial processing, and hydrologic simulation through six integrated components that orchestrate the complete workflow from user prompt to final report.

01 LLM Interface

Converts user prompts into structured simulation instructions specifying locations, time periods, and analytical goals.

02 Tool Executor Layer

Orchestrates Python-based geospatial libraries and model wrappers for comprehensive data retrieval, simulation, and analysis workflows.

03 Dynamic Data Pipeline

Automatically fetches digital elevation models, precipitation, evapotranspiration, and discharge datasets based on user input.

04 Hydrologic Model Integration

Implements the CREST model with dynamically obtained parameters informed by Retrieval-Augmented Generation and LLM reasoning.

05 Report Generation Engine

Compiles simulation outcomes, visualizations, and analytical summaries into publication-quality Markdown or PDF documents.

06 Interactive Feedback Loop

Enables users to refine simulations through natural language, parsing requests, updating configurations, and regenerating reports.

Eight Specialized Agents Orchestrate the Complete Workflow

AQUAH implements a multi-agent system where eight specialized agents collaborate to transform user requests into reproducible simulations. Each agent owns a well-defined responsibility and passes structured artifacts to the next, enabling transparent reasoning, easier debugging, and seamless extensibility.

A_{CP} **Context Parser Agent**

Extracts geographic extent and temporal windows from natural language requests

A_{DR} **Dataset Retriever Agent**

Fetches and clips all required geospatial layers and forcing data

A_{P} **Perceptor Agent**

Employs vision-augmented LLMs to interpret maps and extract morphological descriptors

A_{OS} **OutletSelector Agent**

Applies hydrologic heuristics to identify optimal gauge locations

A_{PI} **ParamInitializer Agent**

Uses Retrieval-Augmented Generation to propose physically plausible initial parameters

A_{O} **Operator Agent**

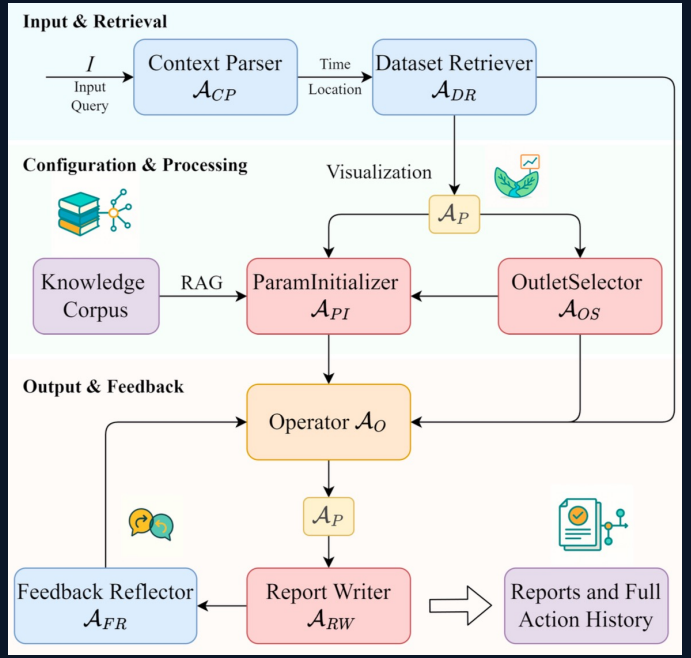
Executes the CREST simulation and captures performance metrics

A_{RW} **Report Writer Agent**

Compiles comprehensive documentation with maps, hydrographs, and statistics

A_{FR} **Feedback Reflector Agent**

Processes user commentary to enable iterative refinement



Automated Data Pipeline Harvests Multi-Source Earth Observations

AQUAH automatically assembles the complete input dataset required for hydrologic simulation without user intervention. Each layer is automatically clipped to the basin envelope and reprojected to a common grid. Missing files trigger fallback notifications and sensible defaults, ensuring workflow robustness across basins with heterogeneous data coverage.

Primary Data Sources

Part 1 of 2

Basin Outlines

Watershed boundary delineation and geographic extent defining the spatial domain for all subsequent data retrieval and model execution

Terrain Products

Digital Elevation Model (DEM), drainage-direction maps, and flow-accumulation maps at 90-meter resolution providing topographic context for hydrologic routing and basin morphology analysis

HydroSHEDS Archive

Precipitation Forcing

High-resolution radar-based precipitation estimates providing the primary meteorological forcing input for water balance calculations and runoff generation

Multi-Radar/Multi-Sensor (MRMS) System

Data Pipeline: Seamless Integration and Quality Assurance

Beyond the primary terrain and precipitation data, AQUAH retrieves Potential Evapotranspiration fields from USGS FEWS-NET for water balance calculations and In-situ Discharge Records from USGS NWIS APIs for streamflow observations and validation. All layers are integrated into a seamless workflow with automatic preprocessing and quality checks, ensuring spatial consistency, temporal alignment, and data completeness.

Automated Integration Workflow

Step 01

Spatial Alignment

All layers are automatically clipped to the basin envelope and reprojected to a common coordinate system and grid resolution

Step 02

Temporal Synchronization

Time series data from multiple sources are aligned to consistent temporal windows with gap detection and interpolation

Step 03

Quality Validation

Automated quality checks verify data completeness, detect anomalies, and trigger fallback mechanisms for missing values

Step 04

Format Conversion

Data are converted to model-compatible formats with appropriate metadata and directory structure for CREST simulation

Step 05

Fallback Handling

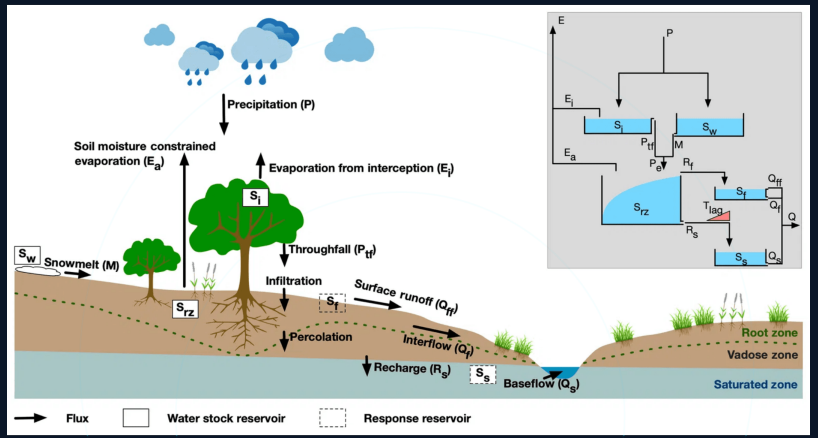
Missing files trigger notifications and sensible default values, ensuring workflow robustness across diverse basins

Step 06

Validation Report

System generates data quality summary documenting sources, coverage, gaps, and preprocessing steps applied

CREST Model Provides Physics-Based Hydrologic Simulation



For runoff generation and routing, AQUAH employs the distributed CREST (Coupled Routing and Excess Storage) model, a well-established physics-based framework for flood simulation and watershed analysis. CREST solves basin water balance components including precipitation partitioning, infiltration, evapotranspiration, and subsurface exchange through coupled differential equations. The model propagates resulting flows using a kinematic-wave routing scheme that accounts for terrain topography and drainage network structure. AQUAH's language-agent modules automatically supply first-guess parameter values informed by basin characteristics, eliminating the traditional need for extensive manual calibration before initial runs.

CREST Model Features

- Distributed water balance calculations
- Precipitation partitioning and infiltration
- Evapotranspiration modeling
- Subsurface exchange processes
- Kinematic-wave flow routing

Multi-Model Evaluation

We asked domain experts to evaluate the report generated by AQUAH, from a variety of metrics: simulation results, reasonableness, clarity

Performance Metrics

Claude-Opus-4	Claude-Sonnet-4	GPT-4o	GPT-o1	Gemini-2.5-Flash
Average Score	Average Score	Average Score	Average Score	Average Score
7.01	6.79	6.30	6.27	6.11

Expert Assessment & Key Findings

- Parameter values consistently achieved correct order of magnitude—far superior to ad-hoc manual guessing
- Hydrologists judged outputs as clear, transparent, and physically plausible
- GPT-4o showed slightly higher median performance across evaluation metrics
- Results suitable for analyst review, though operational deployment requires additional calibration

System Limitations and Future Development Directions

The current AQUAH prototype faces several constraints that future work will address. While initial parameter values prove physically plausible, operational deployment requires additional development to maximize model capabilities and expand geographic coverage.

⚠ Current Limitations

- Single prompt template across all LLMs may not fully leverage each model's unique formatting capabilities or reasoning strengths
- Dependency on publicly hosted data services (USGS, MRMS, cloud-based LLM inference) creates vulnerability to outages
- Geographic coverage currently limited to contiguous United States where comprehensive data availability exists
- Additional calibration and validation workflows required before operational deployment
- Regional access limitations or temporary service outages can reduce system functionality

→ Future Directions

- Model-specific prompt optimization to fully exploit unique capabilities of different LLM architectures
- Local data caching and offline capability to reduce service dependencies and improve reliability
- Global watershed expansion beyond CONUS to support international hydrologic modeling applications
- Automated calibration feedback loops that iteratively refine parameters based on simulation performance
- Integration with additional hydrologic models beyond CREST for broader application domains

Calibration Agent

